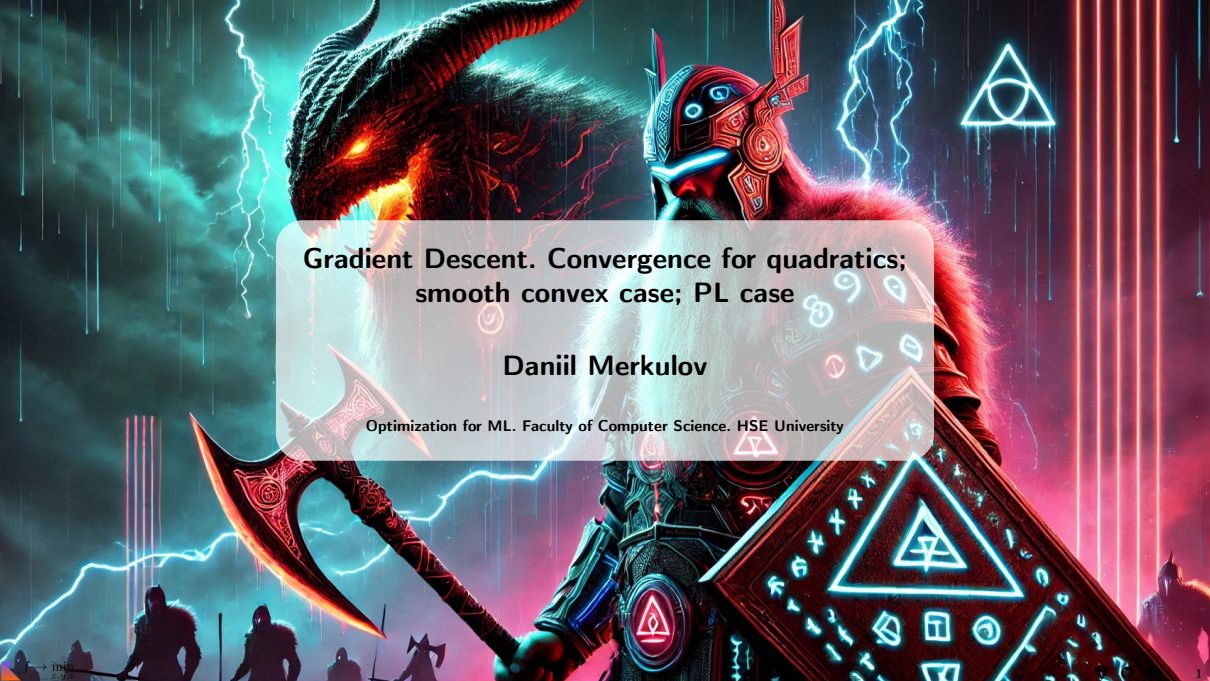

Gradient Descent. Convergence for quadratics; smooth convex case; PL case

Daniil Merkulov

Optimization for ML. Faculty of Computer Science. HSE University

Gradient Descent

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$$x_{k+1} = x_k - \alpha f'(x_k)$$

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Let's consider the following ODE, which is referred to as the Gradient Flow equation.

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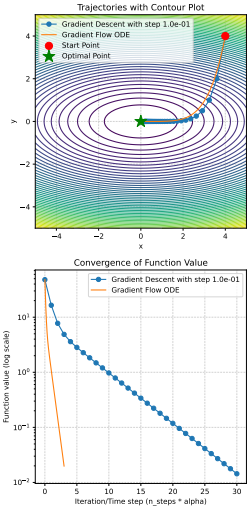
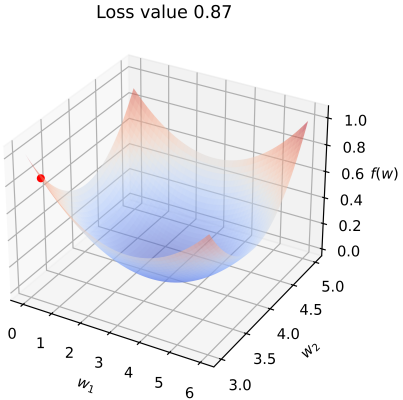
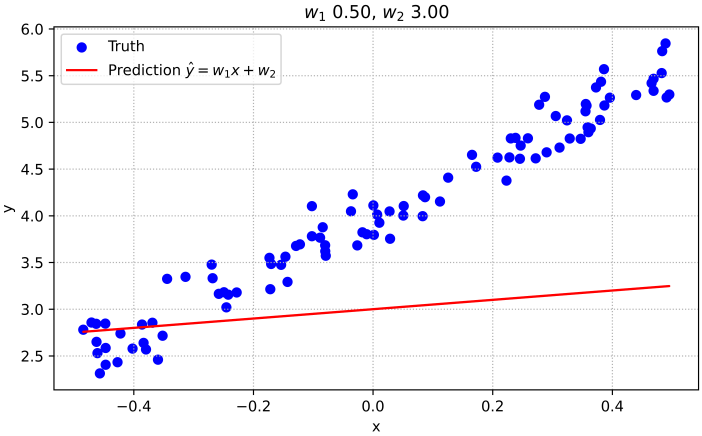


Рисунок 1: Gradient flow trajectory

Convergence of Gradient Descent algorithm

Heavily depends on the choice of the learning rate α:





Exact line search aka steepest descent

$$\alpha_k = \text{arg} \min_{\alpha \in \mathbb{R}^+} f(x_{k+1}) = \text{arg} \min_{\alpha \in \mathbb{R}^+} f(x_k - \alpha \nabla f(x_k))$$

More theoretical than practical approach. It also allows you to analyze the convergence, but often exact line search can be difficult if the function calculation takes too long or costs a lot. An interesting theoretical property of this method is that each following iteration is orthogonal to the previous one:

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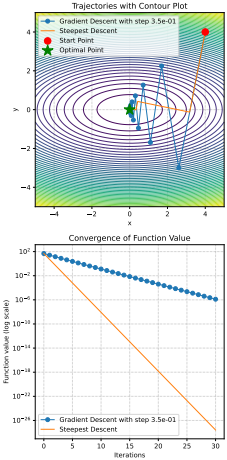

Рисунок 2: Steepest Descent

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Strongly convex quadratics

Coordinate shift

Consider the following quadratic optimization problem:

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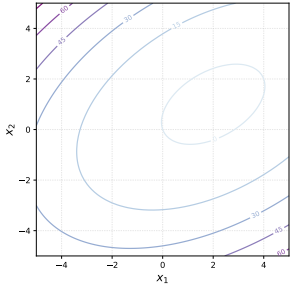

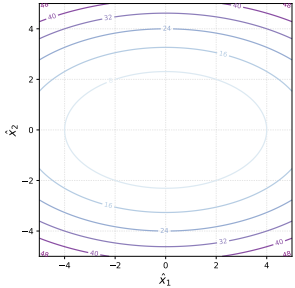

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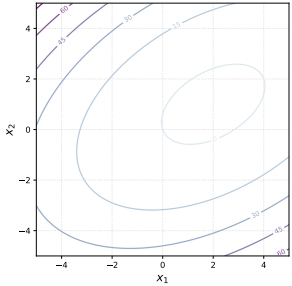

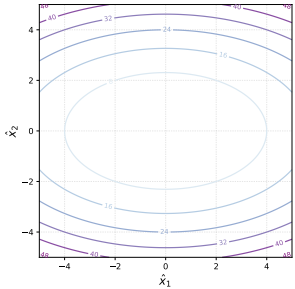

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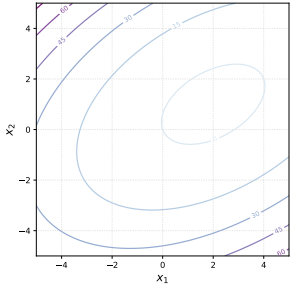

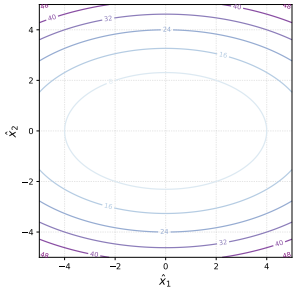

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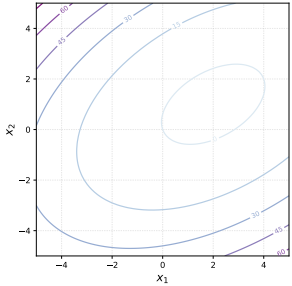

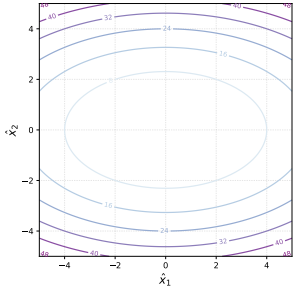

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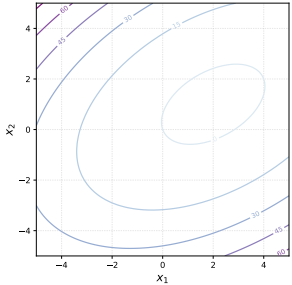

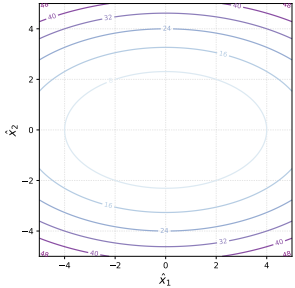

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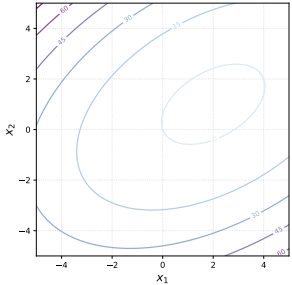

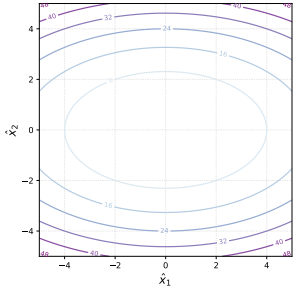

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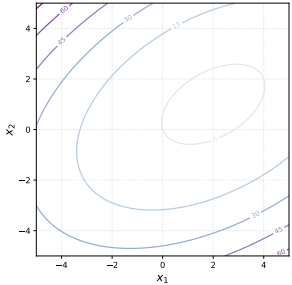

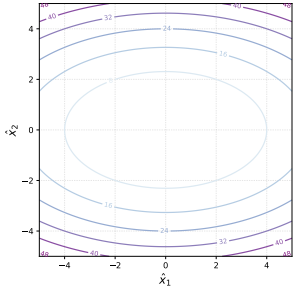

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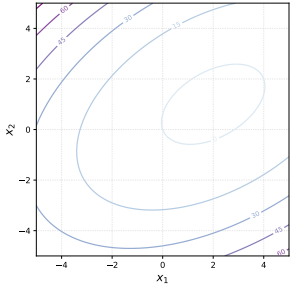

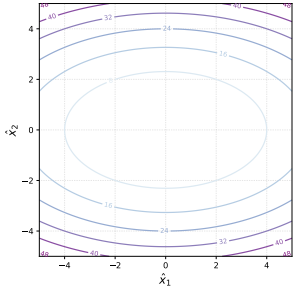

Convergence analysis

Now we can work with the function $f(x) = \frac{1}{2}x^T \Lambda x$ with $x^* = 0$ without loss of generality (drop the hat from the $\hat{x}$)

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$$

Convergence analysis

Now we can work with the function $f(x) = \frac{1}{2}x^T \Lambda x$ with $x^* = 0$ without loss of generality (drop the hat from the $\hat{x}$)

$$
\begin{aligned}
x^{k+1} &= x^k - \alpha^k \nabla f(x^k) = x^k - \alpha^k \Lambda x^k \\
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Let's use constant stepsize $\alpha^k = \alpha$. Convergence condition:

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\rho(\alpha) = \max_i |1 - \alpha \lambda_{(i)}| < 1
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Remember, that $\lambda_{\text{min}} = \mu > 0, \lambda_{\text{max}} = L \geq \mu$.

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Now we would like to tune α to choose the best (lowest) convergence rate

$$
\rho^* = \min_\alpha \rho(\alpha)
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$$

$$
x^{k+1}_{(i)} = \left(\frac{L - \mu}{L + \mu}\right)^k x^0_{(i)}
$$

$$
\|x^{k+1}\|_2 \leq \left(\frac{L - \mu}{L + \mu}\right)^k \|x^0\|_2
$$

Convergence analysis

Now we can work with the function $f(x) = \frac{1}{2}x^T \Lambda x$ with $x^* = 0$ without loss of generality (drop the hat from the $\hat{x}$)

$$
\begin{aligned}
x^{k+1} &= x^k - \alpha^k \nabla f(x^k) = x^k - \alpha^k \Lambda x^k \\
&= (I - \alpha^k \Lambda) x^k \\
x^{k+1}_{(i)} &= (1 - \alpha^k \lambda_{(i)}) x^k_{(i)} \text{ For } i\text{-th coordinate} \\
x^{k+1}_{(i)} &= (1 - \alpha^k \lambda_{(i)})^k x^0_{(i)}
\end{aligned}
$$

Let's use constant stepsize $\alpha^k = \alpha$. Convergence condition:

$$
\rho(\alpha) = \max_i |1 - \alpha \lambda_{(i)}| < 1
$$

Remember, that $\lambda_{\text{min}} = \mu > 0, \lambda_{\text{max}} = L \geq \mu$.

$$
\begin{aligned}
&|1 - \alpha\mu| < 1 \\
&-1 < 1 - \alpha\mu < 1 \\
&\alpha < \frac{2}{\mu} \qquad \alpha\mu > 0
\end{aligned}
\qquad
\begin{aligned}
&|1 - \alpha L| < 1 \\
&-1 < 1 - \alpha L < 1 \\
&\alpha < \frac{2}{L} \qquad \alpha L > 0
\end{aligned}
$$

$\alpha < \frac{2}{L}$ is needed for convergence.

Now we would like to tune α to choose the best (lowest) convergence rate

$$
\begin{aligned}
\rho^* &= \min_\alpha \rho(\alpha) = \min_\alpha \max_i |1 - \alpha \lambda_{(i)}| \\
&= \min_\alpha \{|1 - \alpha\mu|, |1 - \alpha L|\}
\end{aligned}
$$

$$
\alpha^* : \quad 1 - \alpha^* \mu = \alpha^* L - 1
$$

$$
\alpha^* = \frac{2}{\mu + L} \qquad \rho^* = \frac{L - \mu}{L + \mu}
$$

$$
x^{k+1}_{(i)} = \left(\frac{L - \mu}{L + \mu}\right)^k x^0_{(i)}
$$

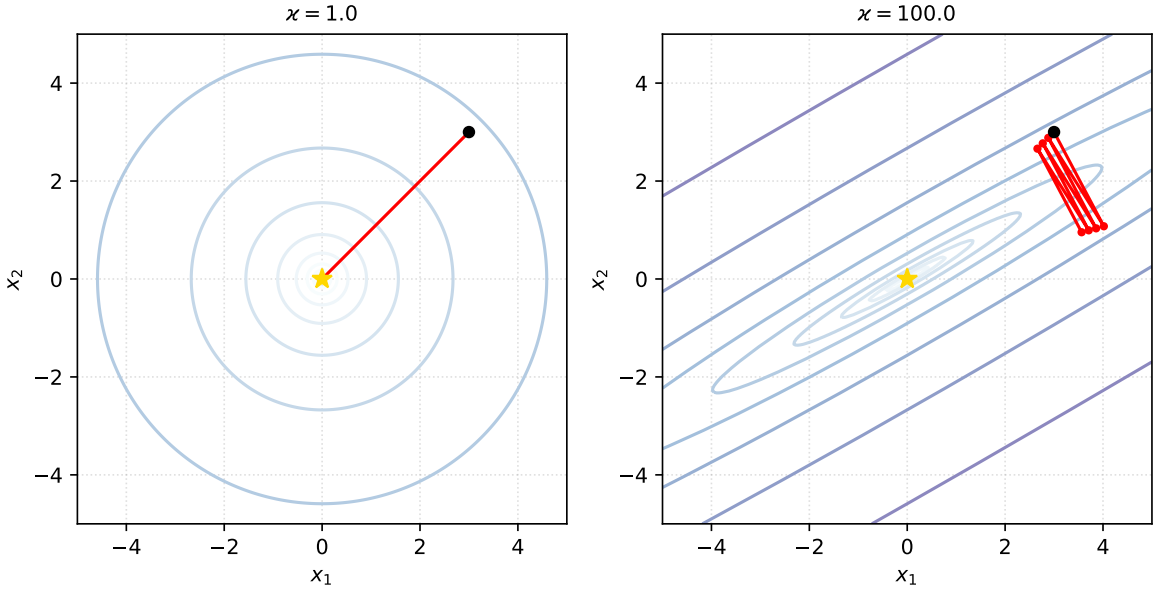
$$
\|x^{k+1}\|_2 \leq \left(\frac{L - \mu}{L + \mu}\right)^k \|x^0\|_2 \quad f(x^{k+1}) \leq \left(\frac{L - \mu}{L + \mu}\right)^{2k} f(x^0
$$

Convergence analysis

So, we have a linear convergence in the domain with rate $\frac{\varkappa-1}{\varkappa+1} = 1 - \frac{2}{\varkappa+1}$, where $\varkappa = \frac{L}{\mu}$ is sometimes called *condition number* of the quadratic problem.

$\varkappa$	ρ	Iterations to decrease domain gap 10 times	Iterations to decrease function gap 10 times
1.1	0.05	1	1
2	0.33	3	2
5	0.67	6	3
10	0.82	12	6
50	0.96	58	29
100	0.98	116	58
500	0.996	576	288
1000	0.998	1152	576

Condition number $\varkappa$



Polyak-Lojasiewicz smooth case

Polyak-Lojasiewicz condition. Linear convergence of gradient descent without convexity

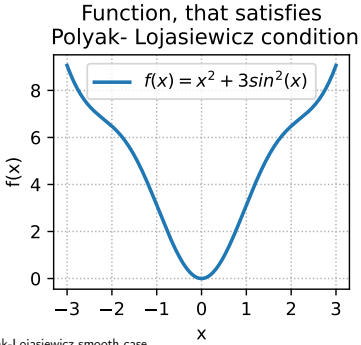
PL inequality holds if the following condition is satisfied for some $\mu > 0$,

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f^*) \quad \forall x$$

It is interesting, that the Gradient Descent algorithm might converge linearly even without convexity.

The following functions satisfy the PL condition but are not convex. Link to the code

$$f(x) = x^2 + 3\sin^2(x)$$



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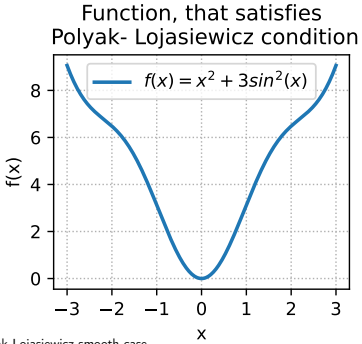
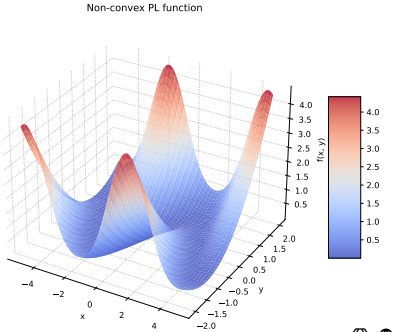
$$f(x) = x^2 + 3\sin^2(x)$$

$$f(x, y) = \frac{(y - \sin x)^2}{2}$$





Convergence analysis

Theorem

Consider the Problem

$$f(x) \to \min_{x \in \mathbb{R}^d}$$

and assume that f is μ-Polyak-Lojasiewicz and L-smooth, for some $L \geq \mu > 0$.
Consider $(x^k)_{k \in \mathbb{N}}$ a sequence generated by the gradient descent constant stepsize algorithm, with a stepsize satisfying $0 < \alpha \leq \frac{1}{L}$. Then:

$$f(x^k) - f^* \leq (1 - \alpha\mu)^k (f(x^0) - f^*).$$

Convergence analysis

We can use L-smoothness, together with the update rule of the algorithm, to write

$$f(x^{k+1}) \leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L}{2} \|x^{k+1} - x^k\|^2$$

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&= f(x^k) - \alpha \|\nabla f(x^k)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x^k)\|^2
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where in the last inequality we used our hypothesis on the stepsize that $\alpha L \leq 1$.

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\end{aligned}
$$

where in the last inequality we used our hypothesis on the stepsize that $\alpha L \leq 1$.

We can now use the Polyak-Lojasiewicz property to write:

$$
f(x^{k+1}) \leq f(x^k) - \alpha\mu(f(x^k) - f^*).
$$

The conclusion follows after subtracting f^* on both sides of this inequality and using recursion.

Any μ-strongly convex differentiable function is a PL-function

Theorem

If a function $f(x)$ is differentiable and μ-strongly convex, then it is a PL function.

Proof

By first order strong convexity criterion:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} \|y - x\|_2^2$$

Putting $y = x^*$:

$$f(x^*) \geq f(x) + \nabla f(x)^T (x^* - x) + \frac{\mu}{2} \|x^* - x\|_2^2$$

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Let $a = \frac{1}{\sqrt{\mu}}\nabla f(x)$ and
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Let $a = \frac{1}{\sqrt{\mu}}\nabla f(x)$ and
$b = \sqrt{\mu}(x - x^*) - \frac{1}{\sqrt{\mu}}\nabla f(x)$
Then $a + b = \sqrt{\mu}(x - x^*)$ and
$a - b = \frac{2}{\sqrt{\mu}}\nabla f(x) - \sqrt{\mu}(x - x^*)$

Any μ-strongly convex differentiable function is a PL-function

$$f(x) - f(x^*) \leq \frac{1}{2}\left(\frac{1}{\mu}\|\nabla f(x)\|_2^2 - \left\|\sqrt{\mu}(x - x^*) - \frac{1}{\sqrt{\mu}}\nabla f(x)\right\|_2^2\right)$$

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$$
f(x) - f(x^*) \leq \frac{1}{2\mu}\|\nabla f(x)\|_2^2,
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f(x) - f(x^*) &\leq \frac{1}{2\mu}\|\nabla f(x)\|_2^2,
\end{aligned}
$$

which is exactly the PL condition. It means, that we already have linear convergence proof for any strongly convex function.

Smooth convex case

Smooth convex case

Theorem

Consider the Problem

$$f(x) \to \min_{x \in \mathbb{R}^d}$$

and assume that f is convex and L-smooth, for some $L > 0$.
Let $(x^k)_{k \in \mathbb{N}}$ be the sequence of iterates generated by the gradient descent constant stepsize algorithm, with a stepsize satisfying $0 < \alpha \leq \frac{1}{L}$. Then, for all $x^* \in \operatorname{argmin} f$, for all $k \in \mathbb{N}$ we have that

$$f(x^k) - f^* \leq \frac{\|x^0 - x^*\|^2}{2\alpha k}.$$

Convergence analysis

- As it was before, we first use smoothness:

$$
\begin{aligned}
f(x^{k+1}) &\leq f(x^k) + \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L}{2}\|x^{k+1} - x^k\|^2 \\
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&\leq f(x^k) - \frac{\alpha}{2}\|\nabla f(x^k)\|^2,
\end{aligned} \tag{1}
$$

$$
f(x^k) - f(x^{k+1}) \geq \frac{1}{2L}\|\nabla f(x^k)\|^2 \text{ if } \alpha = \frac{1}{L}
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Typically, for the convergent gradient descent algorithm the higher the learning rate the faster the convergence. That is why we often will use $\alpha = \frac{1}{L}$.

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- After that we add convexity:

(2)

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f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \tag{2}
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Convergence analysis

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&\leq f(x^k) - \frac{\alpha}{2}\|\nabla f(x^k)\|^2, \\
f(x^k) - f(x^{k+1}) &\geq \frac{1}{2L}\|\nabla f(x^k)\|^2 \text{ if } \alpha = \frac{1}{L}
\end{aligned}
\tag{1}
$$

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$$
\begin{aligned}
f(y) &\geq f(x) + \langle \nabla f(x), y - x \rangle \text{ with } y = x^*, x = x^k \\
f(x^k) - f^* &\leq \langle \nabla f(x^k), x^k - x^* \rangle
\end{aligned}
\tag{2}
$$

Convergence analysis

- Now we put Уравнение 2 to Уравнение 1:

Convergence analysis

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$$f(x^{k+1}) \leq f(x^k) - \frac{\alpha}{2}\|\nabla f(x^k)\|^2 \leq f^* + \langle \nabla f(x^k), x^k - x^* \rangle - \frac{\alpha}{2}\|\nabla f(x^k)\|^2$$

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f(x^{k+1}) &\leq f(x^k) - \frac{\alpha}{2}\|\nabla f(x^k)\|^2 \leq f^* + \langle \nabla f(x^k), x^k - x^* \rangle - \frac{\alpha}{2}\|\nabla f(x^k)\|^2 \\
&= f^* + \langle \nabla f(x^k), x^k - x^* - \frac{\alpha}{2}\nabla f(x^k) \rangle
\end{aligned}
$$

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(3)

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- Due to the monotonic decrease at each iteration $f(x^{i+1}) < f(x^i)$:

$$kf(x^k) \leq \sum_{i=0}^{k-1} f(x^{i+1})$$

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Summary

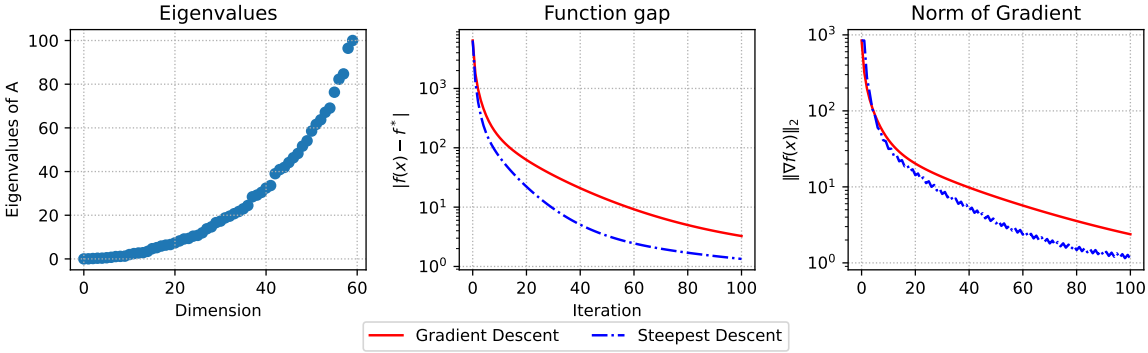
Gradient Descent: $\min\limits_{x \in \mathbb{R}^n} f(x)$ $\quad x^{k+1} = x^k - \alpha^k \nabla f(x^k)$

smooth (non-convex)	smooth & convex	smooth & strongly convex (or PL)
$\|\nabla f(x^k)\|^2 \sim \mathcal{O}\left(\frac{1}{k}\right)$	$f(x^k) - f^* \sim \mathcal{O}\left(\frac{1}{k}\right)$	$\|x^k - x^*\|^2 \sim \mathcal{O}\left(\left(1 - \frac{\mu}{L}\right)^k\right)$
$k_\varepsilon \sim \mathcal{O}\left(\frac{1}{\varepsilon}\right)$	$k_\varepsilon \sim \mathcal{O}\left(\frac{1}{\varepsilon}\right)$	$k_\varepsilon \sim \mathcal{O}\left(\varkappa \log \frac{1}{\varepsilon}\right)$

Numerical experiments

$$f(x) = \frac{1}{2}x^T Ax - b^T x \to \min_{x \in \mathbb{R}^n}$$

Convex quadratics. n=60, random matrix.



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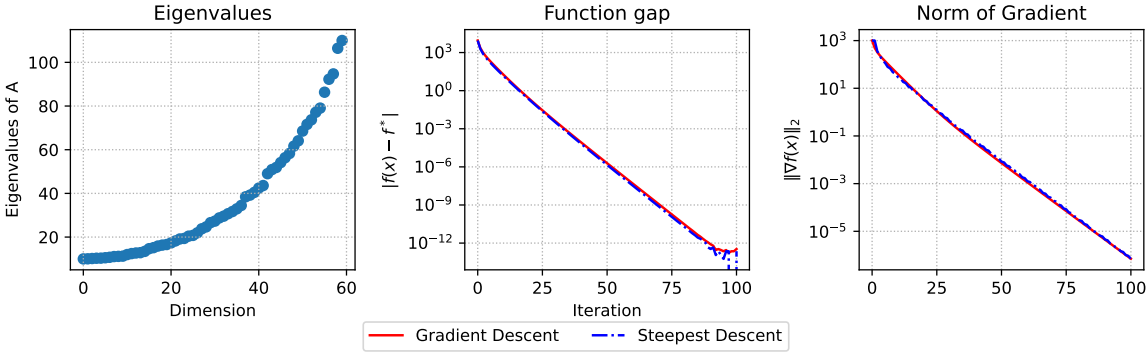

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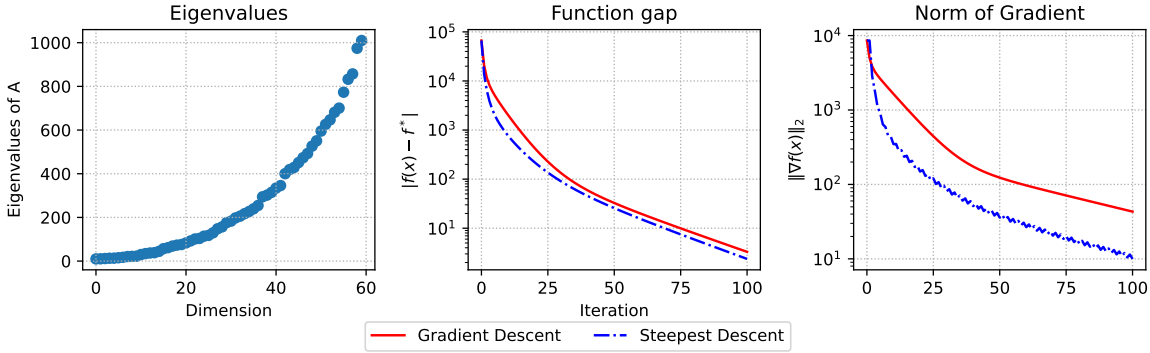

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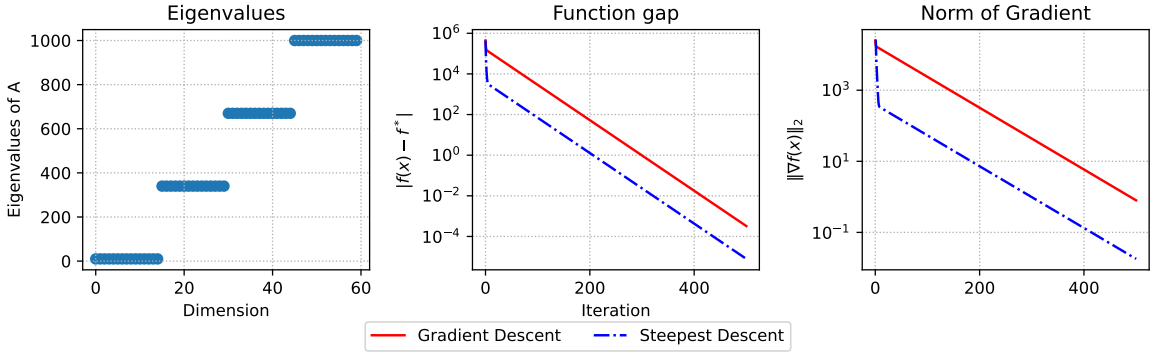

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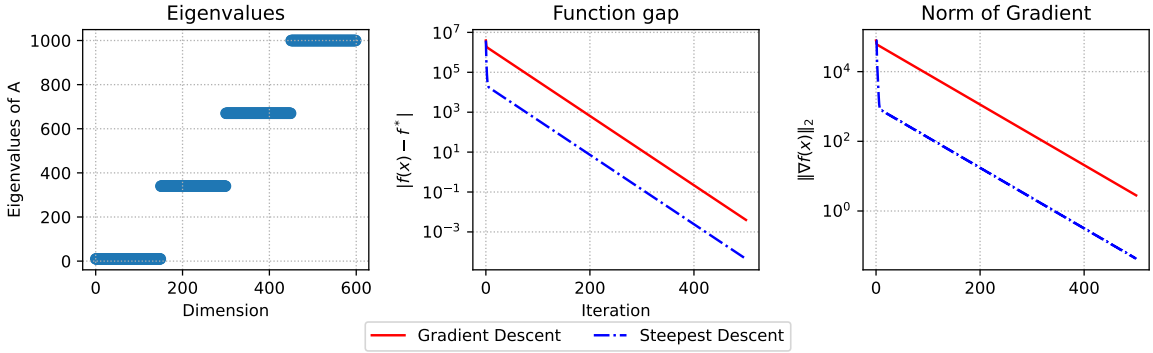
Strongly convex quadratics. n=600, clustered matrix.



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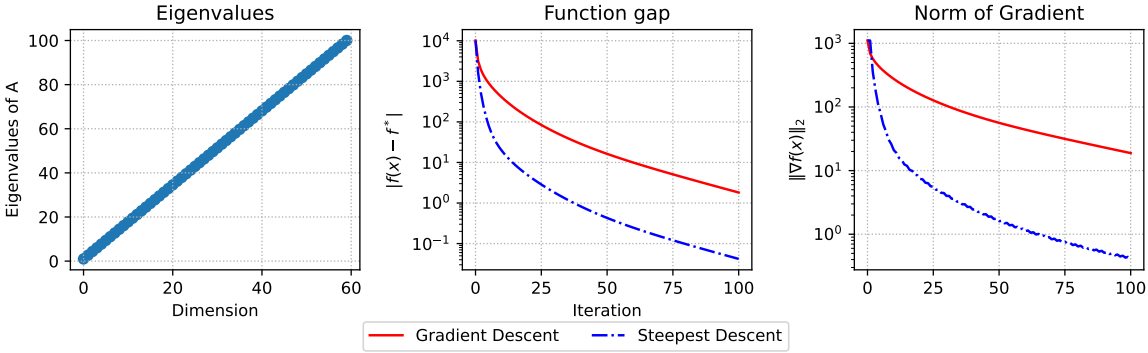
$$f(x) = \frac{1}{2}x^T Ax - b^T x \to \min_{x \in \mathbb{R}^n}$$

Strongly convex quadratics. n=60, uniform spectrum matrix.



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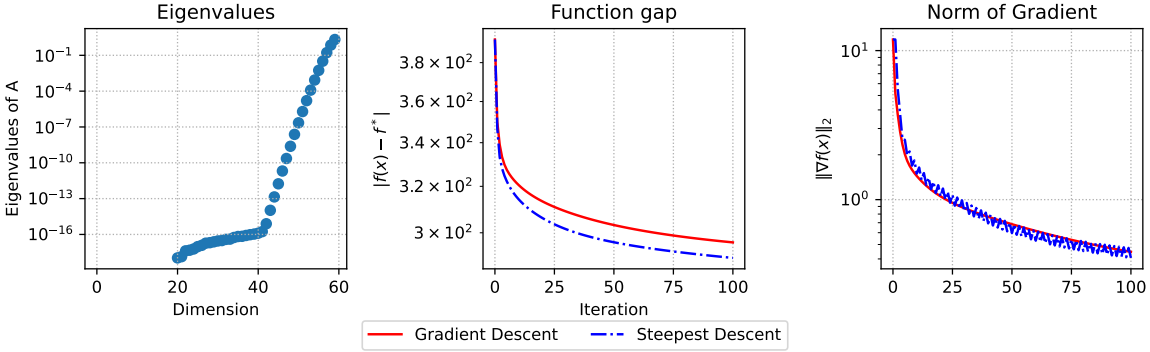
$$f(x) = \frac{1}{2}x^T Ax - b^T x \to \min_{x \in \mathbb{R}^n}$$

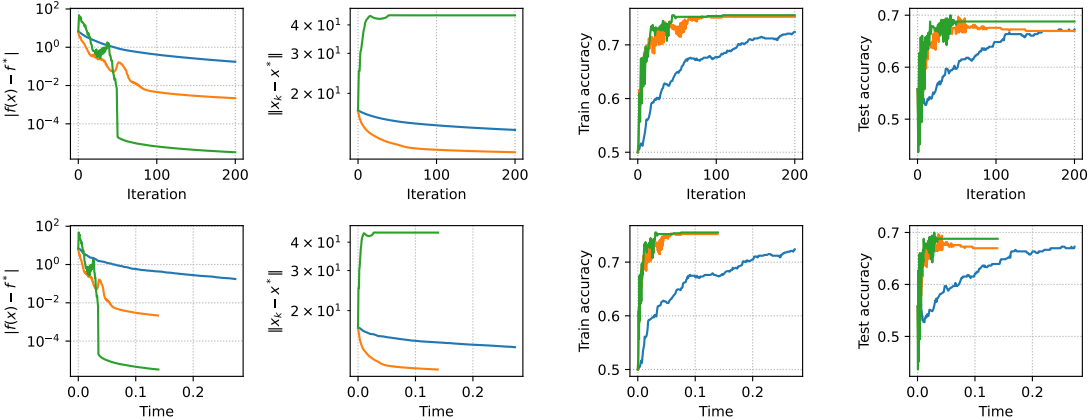
Strongly convex quadratics. n=60, Hilbert matrix.



Numerical experiments

$$f(x) = \frac{\mu}{2}\|x\|_2^2 + \frac{1}{m}\sum_{i=1}^{m}\log(1+\exp(-y_i\langle a_i, x\rangle)) \to \min_{x\in\mathbb{R}^n}$$

Convex binary logistic regression. mu=0.



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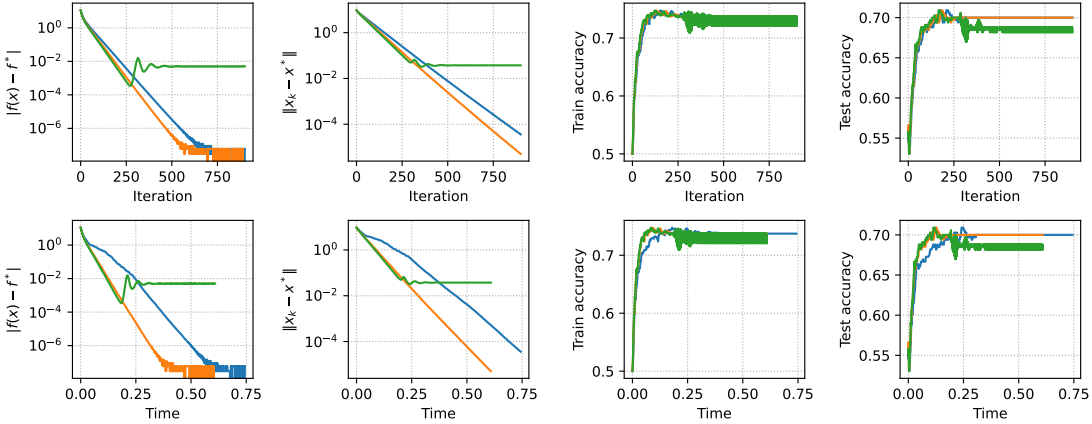
Strongly convex binary logistic regression. mu=0.1.



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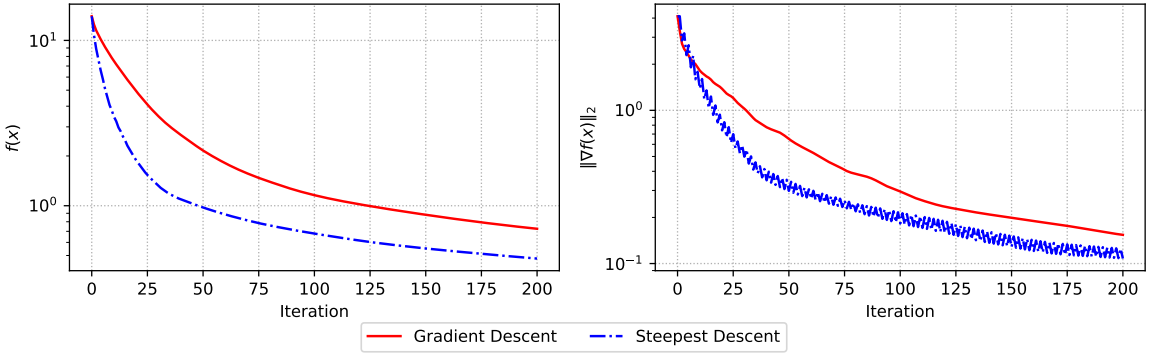
Regularized binary logistic regression. n=300. m=1000. μ=0



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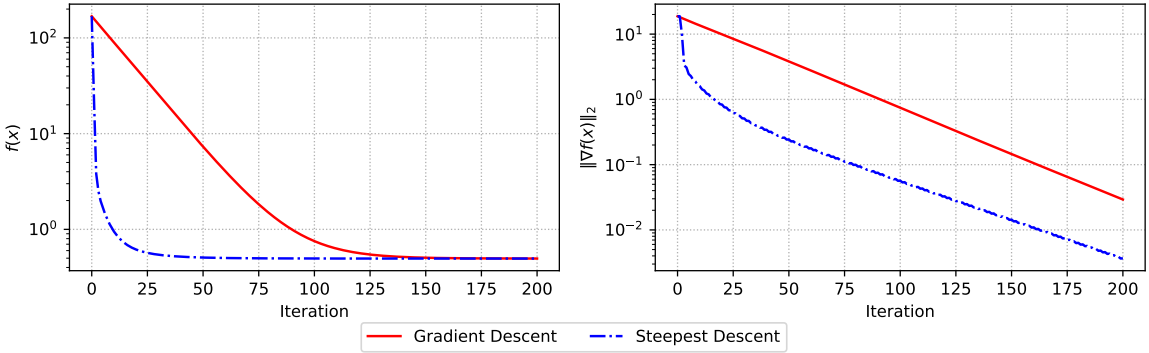
Regularized binary logistic regression. n=300. m=1000. μ=1